FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

FIITJEE **JEE (Main), 2016**

ALL INDIA INTEGRATED TEST SERIES

HALF COURSE TEST – VI

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (04 to 09 and 13 to 30) contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

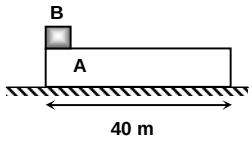
PART – I

SECTION – A

Single Correct Choice Type

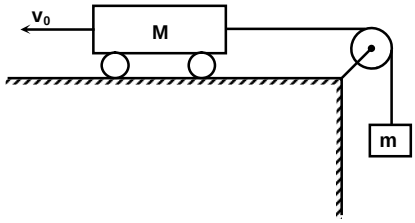
*This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. In the figure shown, a block B of mass 1 kg is placed at the left end of a plank 'A' of mass 4 Kg, which in turn, is placed on a rough horizontal surface. The coefficient of friction between the plank 'A' and the surface is 0.2, whereas friction between the block B and the plank A is absent. Initially both A and B are at rest and at time t = 0, simultaneously they are imparted same velocity v = 10 m/s along the length of the plank. Calculate the time taken by the block to reach the other end of the plank is. (Length of the plank is 40 m).



(A) 6 sec (B) 12 sec
(C) 18 sec (D) 24 sec

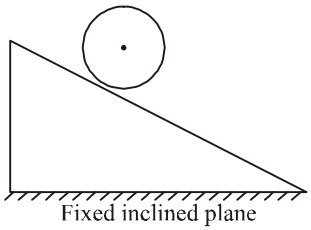
2. A cart with a mass M = $\frac{1}{2}$ kg is connected by a string to a mass of m = 200 g. At the initial moment the cart moves to the left along a horizontal plane at a speed v_0 = 7 m/s. (g = 9.8 m/s^2) Distance covered by cart after 5 sec is



(A) 3.75 (B) 17.5
(C) 20 (D) zero

3. Two identical glass spheres filled with air are connected by a horizontal glass tube. The glass tube contains a pellet of mercury at its mid point. Air in one sphere is at 0°C and the other is at 20°C. If both the vessels are heated through 10°C then neglecting the expansions of the bulbs and the tube. Ignore the conduction of heat by the mercury pellet.
(A) The mercury pellet gets displaced towards the sphere at lower temperature.
(B) The mercury pellet gets displaced towards the sphere at higher temperature.
(C) The mercury pellet does not get displaced at all.
(D) The temperature rise causes the pellet to expand without any displacement.

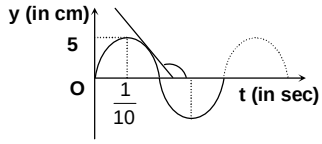
4. An experiment is carried out in which a conducting sphere is kept on the top of smooth inclined plane. Time taken by sphere to reach the bottom is t_1. Now a magnetic field B is switched on parallel to the inclined plane in downward direction and sphere is again left at top of inclined plane. Time taken by sphere in reaching to bottom is t_2 in this case. Now same experiment is repeated with magnetic field B switched on in vertically upward direction. Time taken by sphere in reaching to bottom of inclined plane is t_3 in this case. Then :



(A) $t_1 < t_2 = t_3$ (B) $t_1 = t_2 < t_3$
(C) $t_2 < t_1 < t_3$ (D) $t_1 < t_2 < t_3$

Space for rough work

5. The position of a particle describing simple harmonic motion along y-axis about the origin has been described in the adjacent figure by sinusoidal graph. The maximum speed of the particle is equal to

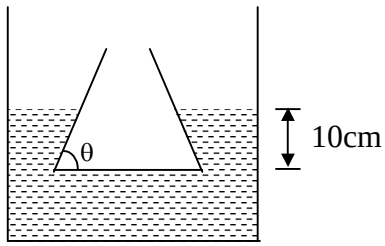


(A) 25π m/s (B) 0.25π m/s
(C) $2.5\ \pi$ m/s (D) none

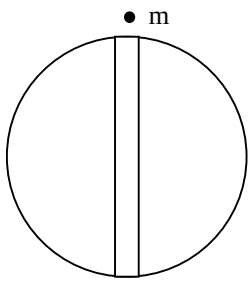
6. A string of length ℓ is fixed at both ends. It is vibrating in its third overtone. Maximum amplitude of the particle on the string is A. The amplitude of the particle at a distance $\ell/3$ from one end is

(A) A (B) 0
(C) $\frac{\sqrt{3}A}{2}$ (D) A/2

7. A conical flask of mass 10 kg and base area as 10^3 cm^2 is floating in liquid of relative density 1.2, as shown in the figure. The force that liquid exerts on curved surface of conical flask will be
(Given g = 10 m/s^2)
(A) 20 N downward direction
(B) 40 N in downward direction
(C) can be determined only if θ is given
(D) 20 N in upward direction

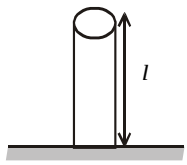


8. A tunnel is made inside earth passing through center of earth. A particle is dropped from the surface of earth. Select the correct statement
(A) Kinetic energy of particle is maximum at center and its potential energy is zero at center
(B) velocity of particle is proportional to x [where x is distance of particle from center of earth
(C) Kinetic energy of particle is maximum when it reaches on the other side of tunnel.
(D) kinetic energy of particle is maximum at center



9. The time period of a pendulum at temperature t_1 °C is T_1 sec. Its time period at t_2°C is T_2 sec. If coefficient of linear expansion of material of pendulum is α, then increase in time period is
(A) $\alpha(t_2 - t_1)$ (B) $\alpha(t_2 - t_1)/3$
(C) $T_1\alpha(t_2 - t_1)/2$ (D) none of these

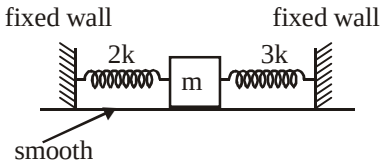
10. A solid cylinder of mass *m* length *l* and area of cross section *A* is placed as shown in the figure. If Young's modulus is *Y* then strain energy stored in the cylinder is



(A) $\frac{m^2g^2l}{3AY}$ (B) $\frac{m^2g^2l}{6AY}$
(C) $\frac{m^2g^2l}{2AY}$ (D) $\frac{m^2g^2l}{AY}$.

Space for rough work

11. A block of mass *m* is connected to two springs of spring constant 2k and 3k respectively, such that the spring of spring constant 2k is compressed by 3*l* when the block is in its mean position. If the block is displaced slightly and allowed to oscillate, the frequency will be



(A) $\frac{1}{2\pi}\sqrt{\frac{k}{m}}$ (B) $\frac{1}{2\pi}\sqrt{\frac{6k}{5m}}$

(C) $\frac{1}{2\pi}\sqrt{\frac{5k}{6m}}$ (D) none of these

12. A wave $y_1 = A\sin\left(2x - 4t + \frac{\pi}{3}\right)$ is superposed with a second waveform producing a standing wave with a node at x = 0. In the equation of the waveform, x is in metre and t is in second. The equation of the second waveform is

(A) $y_2 = A\sin(4t - 2x - \pi/3)$ (B) $y_2 = A\sin(2x + 4t + 5\pi/3)$

(C) $y_2 = A\sin(2x + 4t - 5\pi/3)$ (D) $y_2 = A\sin(-2x + 4t - \pi/3)$.

13. A force $\vec{F} = b\frac{-y\hat{i} + x\hat{j}}{x^2 + y^2}N$ (b is a constant) acts on a particle as it undergoes counterclockwise circular motion in a circle: $x^2 + y^2 = 16$. The work done by the force when the particle undergoes one complete revolution is (x, y are in m)

(A) Zero (B) 2π bJ

(C) 2 bJ (D) None of these.

14. The rate of flow of glycerin of density ρ through conical section of a pipe, if the area of cross-sectional of its ends are A_1 and A_2 and pressure drop across its length be ΔP, is given by

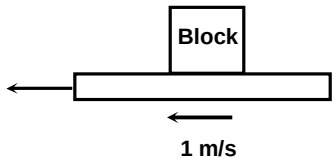
(A) $A_1A_2\sqrt{\frac{\Delta P}{\rho\left(A_1^2 - A_2^2\right)}}$ (B) $A_1A_2\sqrt{\frac{2\Delta P}{\rho\left(A_1^2 - A_2^2\right)}}$

(C) $\sqrt{\frac{2\Delta P}{\rho\left(A_1^2 - A_2^2\right)}}$ (D) $A_1A_2\sqrt{\frac{2\Delta P}{3\rho\left(A_1^2 - A_2^2\right)}}$

Space for rough work

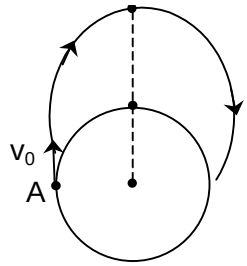
15. For an adiabatic compression (for an ideal gas) the quantity *PV*
(A) increases (B) decreases
(C) remains constant (D) depends on γ

16. A 10 kg block is gently placed on a plank moving at 1 m/s on a frictionless horizontal plane. Due to friction between the plank and the blocks the block accelerates. The velocity of the plank is maintained at a constant 1 m/s by an external force F acting on it. The block finally reaches a velocity of 1 m /s. The work done by the external force, which maintains the constant speed of plank, is [The coefficient of friction between plank and the block is 0.4]. (take g = 10 m/s^2)



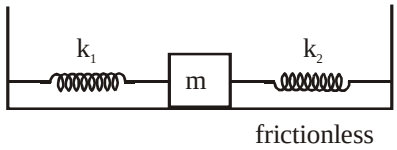
(A) 100 J (B) 50 J
(C) 10 J (D) 5 J.

17. A projectile projected tangentially from the surface of a planet of radius R from a point A, follows the indicated trajectory during its motion. If it is at a height of 3R at the farthest point of its trajectory, then the velocity of projection at A, v_0 is given by (acceleration due to gravity on surface = g)



(A) $v_0 = \sqrt{1.5gR}$ (B) $v_0 = \sqrt{0.5gR}$
(C) $v_0 = \sqrt{1.6gR}$ (D) $v_0 = \sqrt{2gR/3}$

18. Two springs with negligible masses and force constant of $k_1 = 200\ Nm^{-1}$ and $k_2 = 160\ Nm^{-1}$ are attached to the block of mass m = 10 kg as shown in the figure. Initially the block is at rest, at the equilibrium position in which both springs are neither stretched nor compressed. At time t = 0, sharp impulse of 50 N-s is given to the block with a hammer.



(A) period of oscillations for the mass m is $\frac{\pi}{6}$ s
(B) maximum velocity of the mass m during its oscillation is 10 ms^{-1}
(C) data are insufficient to determine maximum velocity
(D) amplitude of oscillation is 5/6 m

19. 1 mole of H_2 and 1 mole of He at a pressure of P_0 & $2P_0$ respectively and a temperature T_0 are kept in two different containers. The ratio of their rms velocities is $C_{H_2} : C_{He}$ =

(A) $\sqrt{10} : \sqrt{3}$ (B) 1 : 1
(C) 5 : 3 (D) 2 : $\sqrt{2}$

Space for rough work

20. An equimolar mixture of a monoatomic and a diatomic ideal gas is suddenly compressed to $\frac{1}{8}^{th}$ of its original volume. The ratio of final temperature to the initial temperature.
(A) $8^{0.53}$ (B) $8^{0.5}$
(C) $8^{1.53}$ (D) $8^{2/3}$

21. An insect of mass m is initially at one end of the stick of length L and mass M, which rests on a smooth horizontal floor. The coefficient of friction between the insect and stick is k. The minimum time in which the insect can reach the other end of the stick is t. Mark all the correct statements.
(A) the centre of mass of the plank has velocity magnitudes $\left[\frac{kmg}{M}\right]t$ w.r.t. horizontal floor at time t.
(B) $t = 2\sqrt{\frac{LM}{k(M+m)g}}$
(C) the magnitude of the linear momentum of the insect at time 't' is (kmgt) w.r.t. horizontal floor
(D) all the above

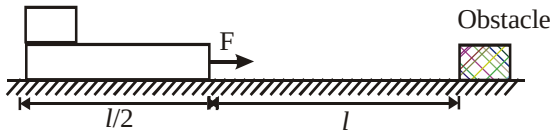
22. A trolley is moving horizontally with a velocity of v m/s w.r.t. earth. A man starts running from one end of the trolley with a velocity 1.5 v m/s w.r.t. the trolley. After reaching the opposite end, the man turns back and continues running with a velocity of 1.5 v m/s w.r.t. trolley in the backward direction. If the length of the trolley is L then the displacement of the man with respect to earth measured as a function of time, will attain a maximum value of
(A) $\frac{4}{3}L$ (B) $\frac{2}{3}L$
(C) $\frac{5L}{3}$ (D) 1.5 L

23. The molar heat capacity of a diatomic ideal gas undergoing the process : $PV^{1/4}$ = constant is
(A) $\frac{17}{6}R$ (B) $\frac{2}{5}R$
(C) $\frac{23}{6}R$ (D) 0

24. An organ pipe P_1 closed at one end vibrating in its first harmonic and another pipe P_2 open at both ends vibrating in its third harmonic are in resonance with a given tuning fork. The ratio of the length of P_1 to that of P_2 is
(A) $\frac{8}{3}$ (B) $\frac{3}{8}$
(C) $\frac{1}{6}$ (D) $\frac{1}{3}$

Space for rough work

25. As shown in the diagram, there are two blocks of the same mass M, one on top of the other, lying on a friction less horizontal surface. Both the blocks are at rest. The upper block is much smaller than the lower block. A force F is applied on the lower block and both the blocks start moving together without any relative motion. Suddenly, the lower block hits a fixed obstacle and comes to rest. The upper block continues to slide on the lower block. The upper block just manages to reach the opposite end of the lower block. What is the coefficient of friction between the two blocks ?

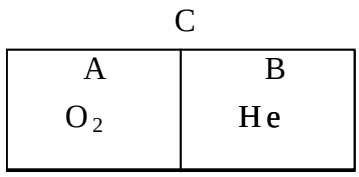


(A) F/Mg
(B) 2F/Mg
(C) F/2Mg
(D) none

26. Two point masses A of mass M and B of mass 4M are fixed at the ends of a rod of length ℓ and of negligible mass. The rod is set rotating about an axis perpendicular to its length with a uniform angular speed. For the rod to have the minimum possible kinetic energy of rotation, the axis of rotation should be at a distance of

(A) $\frac{2}{5}\ell$ from A
(B) $\frac{8}{5}\ell$ from A
(C) $\frac{4}{5}\ell$ from A
(D) $\frac{\ell}{5}$ from A

27. A fixed chamber isolated from surrounding is divided into two equal half A and B as shown in the figure. Part A contains one mole of oxygen and Part B contains one mole of helium, the separator C is thermally conducting and kept fixed. Initial temperature of oxygen chamber is 600 K and that of helium 300 K. Specific heat capacity of separator and chamber is negligible. Choose the correct statement.



(A) change of temperature of helium is equal to change in temperature of oxygen
(B) change in internal energy of helium is equal to 200 R joule (R represents the value of the gas constant in SI)
(C) temperature of oxygen in steady state condition is 487 K (approx)
(D) temperature in steady state can't be calculated

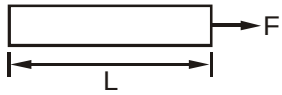
Space for rough work

28. A shot whose mass is m penetrates a thickness S of a fixed plate of mass M. If M were free to move the thickness penetrated will be

(A) $\frac{SM}{m}$ (B) $\frac{Sm}{(M+m)}$

(C) $\frac{SM}{(M+m)}$ (D) $\frac{Sm}{M}$.

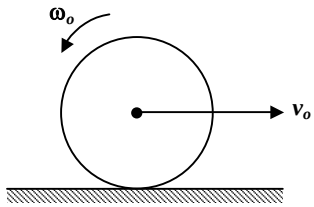
29. A rod of length *L* is pulled with a force *F* as shown on a smooth horizontal surface. If *A* is the area of cross-section and *Y* the Young's modulus of the material of the rod, the elastic potential energy stored in the rod is



(A) $\frac{F^2L}{3YA}$ (B) $\frac{F^2L}{2YA}$

(C) $\frac{F^2L}{6YA}$ (D) $\frac{F^2L}{YA}$

30. A uniform circular disc of radius *r* placed on a rough horizontal surface has initially a velocity v_o and angular velocity ω_o as shown in the figure. The disc comes to rest completely after moving some distance. Then $\frac{v_o}{r\omega_o}$ is



(A) 1 (B) 1/2

(C) 3/2 (D) 2

Space for rough work

Chemistry

PART – II

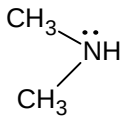
SECTION – A

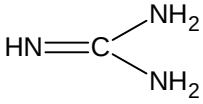
Straight Objective Type

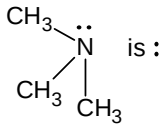
*This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

1. The order of basic character of

(I) $(CCl_3)_3\ddot{N}$

(II) 

(III) 

(IV)  is :

(A) I > III > II > IV
(B) III > II > IV > I
(C) I > II > IV > III
(D) IV > III > II > I

2. Arrange the following oxides in decreasing acidic order:

SiO_2	CO_2	SO_3	N_2O_5
(I)	(II)	(III)	(IV)

(A) III > IV > II > I
(B) IV > III > II > I
(C) IV > II > III > I
(D) III > II > IV > I

3.

List – 1 (ionic species)	**List – 2 (shapes)**
(P) XeF_5^+	(1) Tetrahedral
(Q) SiF_5^-	(2) Square planar
(R) AsF_4^+	(3) Trigonal bipyramidal
(S) ICl_4^-	(4) Square pyramidal
	(5) Octahedral

	P	**Q**	**R**	**S**
(A)	1	2	5	3
(C)	4	2	1	3

	P	**Q**	**R**	**S**
(B)	1	3	5	2
(D)	4	3	1	2

Space for rough work

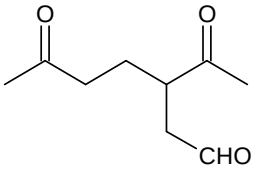
4. The radial probability curve obtained for an orbital wave function (ψ) has 3 peaks and 2 radial nodes. The valence electron of which one of the following metals does this wave function (ψ) correspond to?
(A) Ca (B) Li
(C) K (D) Na

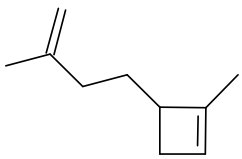
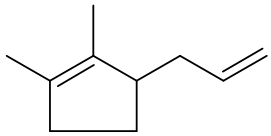
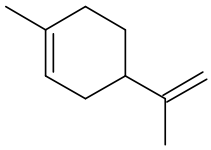
5. What is the minimum pH required to prevent the precipitation of ZnS in a solution i.e. 0.01 M $ZnCl_2$ and saturated with 0.1 M H_2S?
$K_{sp(ZnS)} = 10^{-21}$; $K_{a_1(H_2S)} \times K_{a_2(H_2S)} = 10^{-20}$
(A) 4 (B) 1
(C) 3 (D) 2

6. The relative rates of effusion of O_2 to CH_4 through a container containing O_2 and CH_4 in 3:2 mass ratio will be
(A) $\frac{3\sqrt{2}}{4}$ (B) $\frac{3}{4\sqrt{2}}$
(C) $\frac{3}{2\sqrt{2}}$ (D) none of these

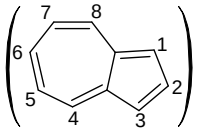
7. After 280 days the activity of a radioactive sample is 6000 dps, the activity reduces to 3000 dps after another 140 days. The initial activity of the sample in dps is
(A) 6000 (B) 9000
(C) 3000 (D) 2400

8. Compound (A) on hydrogenation consumed 2 mole equivalents of hydrogen and produces a saturated hydrocarbon. When A is treated with ozone, followed by reductive work up two products are isolated, formaldehyde and the diketone, structure of which is given below



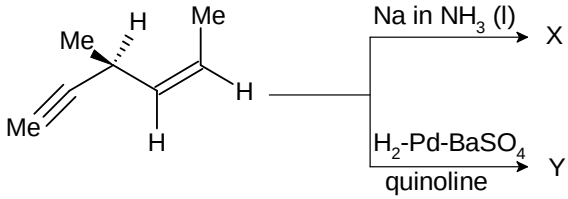
Compound (A) can be represented as
(A) 
(B) 
(C) 
(D) All of these

Space for rough work

9. When azulene 

is treated with electrophile, it attacks which position?

(A) position –1
(B) position 1 & 3
(C) position – 4
(D) position 1 & 4

10. 

The correct statements regarding X and Y is/are
(A) both X and Y are optically inactive
(B) both X and Y are optically active
(C) X is optically inactive where as Y is inactive
(D) X is optically active where are Y is inactive

11. Cl_2 gas has been introduced in an 8 litres cubical container at pressure of 5 atm and at 300 K temperature. The following equilibrium has been achieved

$$Al_{(s)} + \frac{3}{2}Cl_{2(g)} \rightleftharpoons AlCl_{3(s)}$$

If the thickness of inner side of walls has been decreased by 0.04 mm at equilibrium, then find out K_p for the reaction ($\rho_{Al} = 1.13$ g / cc)

(A) $(2.7)^{\frac{3}{2}}$
(B) $(2.7)^{-\frac{3}{2}}$
(C) $(5.4)^{\frac{3}{2}}$
(D) $(5.4)^{-\frac{3}{2}}$

12. For the reaction at 27°C, in 5 litre container $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, ΔH = 5 KJ/mole. Kp of the reaction can be calculated from the following equation Kp = $\frac{bcRT}{aV}$ where b, c and a are equilibrium moles of PCl_3, Cl_2 and PCl_5 respectively. On increasing temperature
(A) dissociation of PCl_5 will increase
(B) dissociation of PCl_5 will decrease
(C) pressure of container will increase
(D) both (A) and (C)

Space for rough work

13. $B(OH)_3 + NaOH \rightleftharpoons Na[B(OH)_4]$

How can this reaction be made to proceed in forward direction by?

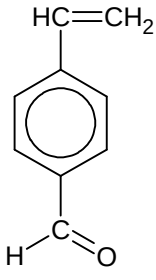
(A) Addition of borax
(B) Addition of cis-1,2-diol
(C) Addition of trans-1,2-diol
(D) Addition Na_2HPO_4

14. The aqueous solution of unknown salt of sodium give the following reactions

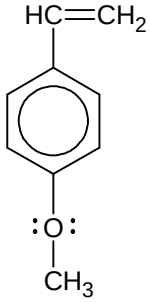
(i) It gives white turbidity with dilute HCl
(ii) It decolorizes a solution of iodine in potassium iodide
(iii) It gives white ppt. with $AgNO_3$ solution which changes colours and finally becomes black on standing

(A) sodium sulphite
(B) sodium sulphide
(C) sodium bisulphate
(D) sodium thiosulphide

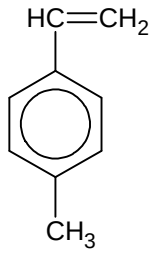
15. Arrange the following compounds in decreasing order of rate of electrophilic addition reaction:



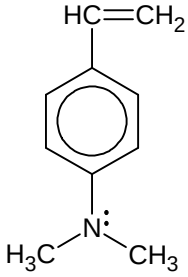
(I)



(II)



(III)

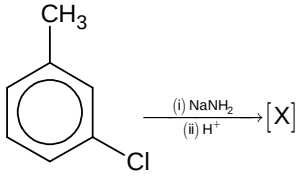


(IV)

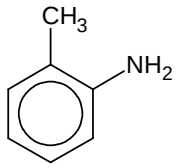
(A) II > IV > I > III
(B) IV > II > III > I
(C) I > II > III > IV
(D) III > II> IV > I

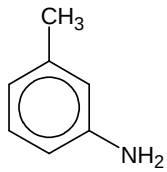
Space for rough work

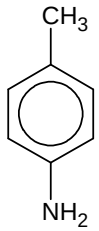
16. Consider the following reaction:

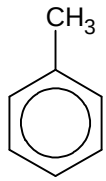


[X] will be

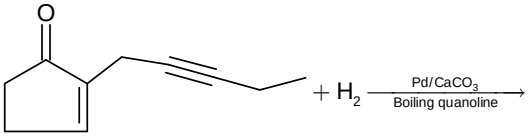
(A) 

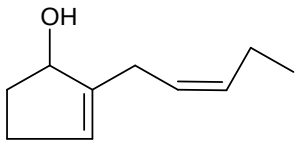
(B) 

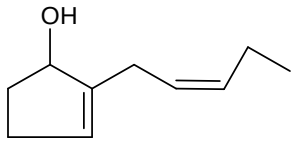
(C) 

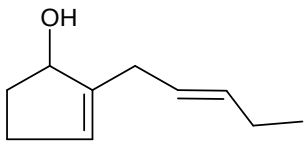
(D) 

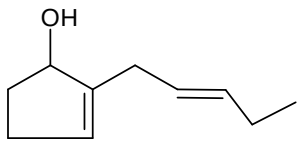
17. The product of the following reaction is



(A) 

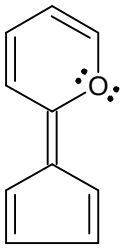
(B) 

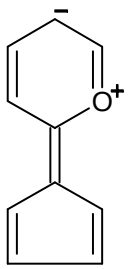
(C) 

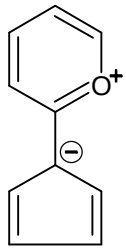
(D) 

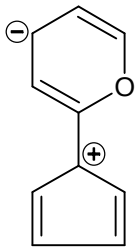
Space for rough work

18. The most stable canonical structures of this molecule is



(A) 

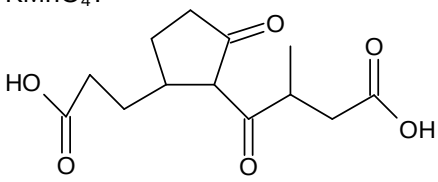
(B) 

(C) 

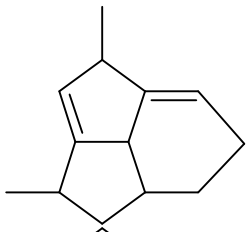
(D) None of these

Space for rough work

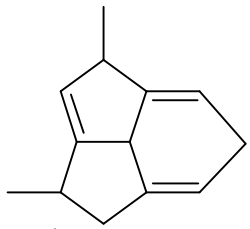
19. Which molecule will give following dicarboxylic acid upon treatment with acidic solution of $KMnO_4$?



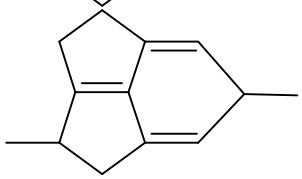
(A)



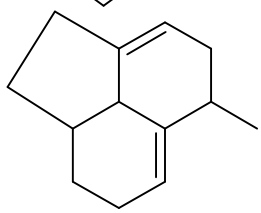
(B)



(C)



(D)



20. Among LiCl, RbCl, $BeCl_2$, $MgCl_2$ the compounds with greatest and least ionic character respectively are
(A) LiCl, RbCl
(B) RbCl, $BeCl_2$
(C) RbCl, $MgCl_2$
(D) $MgCl_2$, $BeCl_2$

21. Which of the following carbides give allylene on hydrolysis?
(A) CaC_2
(B) Be_2C
(C) Mg_2C_3
(D) MgC_2

Space for rough work

22. Mortar is a mixture of
(A) $Ca(OH)_2$, silica and water
(B) $CaCO_3$ and SiO_2
(C) CaO and silica
(D) $CaCO_3$, SiO_2 and water

23. Beryllium show diagonal relationship with aluminium. Which one of the following properties is not shown by both?
(A) BeO and Al_2O_3 amphoteric
(B) Be_2C and Al_4C_3 are methanides
(C) Both have same maximum covalency
(D) Both occur together in nature in beryl

24. At 25°C, the value of pk_b for NH_3 in aqueous solution is 4.7. What is the pH of 0.1 M aqueous solution of NH_4Cl with 0.01 N NH_3?
(A) 8.3
(B) 9
(C) 9.5
(D) 10

25. Equal volume of following solutions are mixed. In which case the pH of resulting solution will be average pH value of two solutions?
Given $K_{a\ (HCN)} = 10^{-10}$, $K_{a\ (CH_3COOH)} = K_{b\ (NH_3)_{aq.}} = 1.8 \times 10^{-5}$
(A) HCl (pH = 2) and NaOH (pH = 12)
(B) HCl (pH = 2) and HCl (pH = 4)
(C) HCN (pH = 2) and NaOH (pH = 12)
(D) CH_3COOH (pH = 6) and NH_3(aq)(pH = 9)

26. In a titration of a weak acid HX with NaOH. The pH is 5.8 after the addition of 10 mL of NaOH and is 6.8 after the addition of 20 mL of NaOH solution. Then ionization constant of acid is
[log 2 = 0.3 and log 5 = 0. 7]
(A) 5.9
(B) 6.8
(C) 7.2
(D) 4.8

Space for rough work

27. Standard free energy changes at 2000 K are as shown below:

Reaction	ΔG^o (kJ mol^{-1})
$2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$	–1140
$2Ni(s) + O_2(g) \longrightarrow 2NiO(s)$	–432
$2Zn(s) + O_2(g) \longrightarrow 2ZnO(s)$	–696

At 200 K, which of the following is/are feasible?

(A) Reduction of MgO by Zn
(B) Reduction of zinc oxide by nickel
(C) Oxidation of magnesium by nickel oxide
(D) Oxidation of nickel by zinc oxide

28. An ideal gas with pressure P, volume V and temperature T is expanded isothermally to a volume 2V and a final pressure P_i. If the same gas is expanded adiabatically to a volume 2V, the final pressure is P_a. The ratio of the specific heat for the gas is 1.67. The ratio of P_a/P_i is:

(A) $\frac{1}{(2)^{0.67}}$
(B) $(2)^{0.67}$
(C) $\left(-\frac{1}{2}\right)^{-0.67}$
(D) $\frac{1}{2^{-0.67}}$

29. Hydrogen [H], deuterium (D) and singly ionized helium (He^+) all have one electron around the nucleus. The wavelengths of radiations emitted for their transitions from n = 2 to 1 are λ_1, λ_2 and λ_3 respectively. Then approximately:

(A) $4\lambda_1 = 2\lambda_2 = 2\lambda_3$
(B) $\lambda_1 = 2\lambda_2 = 2\sqrt{2}\lambda_3$
(C) $\lambda_1 = \lambda_2 = 2\lambda_3$
(D) $\lambda_1 = \lambda_2 = 4\lambda_3$

30. If w_1, w_2, w_3 and w_4 for an ideal gas are work done in isothermal, adiabatic, isobaric and isochoric reversible expansion processes, the incorrect order is / are:

(A) $w_1 > w_2$
(B) $w_3 > w_2$
(C) $w_1 > w_2 > w_3$
(D) $w_3 > w_4$

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

*This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

1. The circle $x^2 + y^2 = 4$ cuts the line joining the points A(1, 0) and B(3, 4) in two points P and Q. Let $\frac{BP}{PA} = \alpha$ and $\frac{BQ}{QA} = \beta$. Then α and β are roots of the quadratic equation

   (A) $3x^2 + 2x - 21 = 0$ (B) $3x^2 + 2x + 21 = 0$

   (C) $2x^2 + 3x - 21 = 0$ (D) None of these

2. The eccentricity of the ellipse $ax^2 + by^2 + 2fx + 2gy + c = 0$ if axis of ellipse parallel to x-axis is

   (A) $\sqrt{\left(\frac{b-a}{b}\right)}$ (B) $\sqrt{\left(\frac{a+b}{b}\right)}$

   (C) $\sqrt{\left(\frac{a+b}{a}\right)}$ (D) none of these

3. If $\frac{(1-3x)^{1/2} + (1-x)^{5/3}}{\sqrt{4-x}}$ is approximately equal to $a + bx$ for small values of x, then $(a,b) =$

   (A) $\left(1, \frac{35}{24}\right)$ (B) $\left(1, -\frac{35}{24}\right)$

   (C) $\left(2, \frac{35}{12}\right)$ (D) $\left(2, -\frac{35}{12}\right)$

Space for rough work

4. A pair of straight lines drawn through the origin form with the line $2x+3y=6$ an isosceles right angled triangle, then the lines and the area of the triangle thus formed is

(A) $x-5y=0$, $5x+y=0$, $\Delta=\frac{36}{13}$

(B) $3x-y=0$, $x+3y=0$, $\Delta=\frac{12}{17}$

(C) $5x-y=0$, $x+5y=0$, $\Delta=\frac{13}{5}$

(D) None of these

5. The vertices of a triangle are (2, 1), (5, 2) and (4, 4). The lengths of the perpendicular from these vertices on the opposite sides are

(A) $\frac{7}{\sqrt{5}},\frac{7}{\sqrt{13}},\frac{7}{\sqrt{6}}$

(B) $\frac{7}{\sqrt{6}},\frac{7}{\sqrt{8}},\frac{7}{\sqrt{10}}$

(C) $\frac{7}{\sqrt{5}},\frac{7}{\sqrt{8}},\frac{7}{\sqrt{15}}$

(D) $\frac{7}{\sqrt{5}},\frac{7}{\sqrt{13}},\frac{7}{\sqrt{10}}$

6. If the straight line through the point $P(3,4)$ makes an angle $\frac{\pi}{6}$ with the x-axis and meets the line $12x+5y+10=0$ at Q, then the length PQ is

(A) $\frac{132}{12\sqrt{3}+5}$

(B) $\frac{132}{12\sqrt{3}-5}$

(C) $\frac{132}{5\sqrt{3}+12}$

(D) $\frac{132}{5\sqrt{3}-12}$

7. The locus of a point which divides the join of $A(-1,1)$ and a variable point P on the circle $x^2+y^2=4$ in the ratio 3 : 2, is

(A) $25(x^2+y^2)+20(x-y)+28=0$

(B) $25(x^2+y^2)+20(x-y)-28=0$

(C) $20(x^2+y^2)+25(x-y)+28=0$

(D) None of these

Space for rough work

8. The equation of the latusrectum of the parabola $y^2 - 6y + 4x - 3 = 0$ is
(A) $x = -2$ (B) $x = 2$
(C) $x = -3/2$ (D) $x = 3/2$

9. The curve described parametrically by $x = t^2 + t + 1,\ y = t^2 - t + 1$ represents
(A) a circle (B) a parabola
(C) an ellipse (D) a hyperbola

10. In a certain test there are n questions. In the test 2^{n-i} students gave wrong answers to at least i questions, where $i = 1, 2, ...n$. If the total number of wrong answers given is 2047, then n is equal to
(A) 10 (B) 11
(C) 12 (D) 13

11. If $\tan x = \frac{2b}{a-c}(a \neq c)$,
$y = a\cos^2 x + 2b\sin x\cos x + c\sin^2 x$
and $z = a\sin^2 x - 2b\sin x\cos x + c\cos^2 x$, then
(A) $y = z$ (B) $y + z = a + c$
(C) $y - z = a + c$ (D) $y - z = (a-c)^2 + 4b^2$

12. Given that the equation $z^2 + (p + iq)z + r + is = 0$, where p, q, r, s are real and non-zero has a real root, then
(A) $pqr = r^2 + p^2 s$ (B) $prs = q^2 + r^2 p$
(C) $qrs = p^2 + s^2 q$ (D) $pqs = s^2 + q^2 r$

13. The eccentricity of the conic $x = 3\left(\frac{1-t^2}{1+t^2}\right)$ and $y = \frac{2t}{1+t^2}$ is
(A) $\frac{2}{3}$ (B) $\frac{2\sqrt{2}}{3}$
(C) $\frac{3}{2}$ (D) $\frac{3}{2\sqrt{2}}$

Space for rough work

14. If P = (x, y), F_1 = (3, 0), F_2 = (–3, 0) and $16x^2 + 25y^2 = 400$, then $PF_1 + PF_2$ equals
(A) 8 (B) 6
(C) 10 (D) 12

15. A hyperbola, having the transverse axis of length $2\sin\theta$, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then its equation is:
(A) $x^2\,\text{cosec}^2\theta - y^2\sec^2\theta = 1$ (B) $x^2\sec^2\theta - y^2\text{cosec}^2\theta = 1$
(C) $x^2\sin^2\theta - y^2\cos^2\theta = 1$ (D) $x^2\cos^2\theta - y^2\sin^2\theta = 1$

16. If the coefficient of the middle term in the expansion of $(1+x)^{2n+2}$ is p and the coefficients of middle terms in the expansion of $(1+x)^{2n+1}$ are q and r, then
(A) $p + q = r$ (B) $p + r = q$
(C) $p = q + r$ (D) $p + q + r = 0$

17. The number of ways of dividing 52 cards amongst four players so that three players have 17 cards each and the fourth player just one card, is
(A) $\frac{52!}{(17!)^3}$ (B) $52!$
(C) $\frac{52!}{17!}$ (D) None of these

18. If $\tan(\cot x) = \cot(\tan x)$, then $\sin 2x$ =
(A) $(2n+1)\frac{\pi}{4}$ (B) $\frac{4}{(2n+1)\pi}$
(C) $4\pi(2n+1)$ (D) None of these

19. If $1 + \sin x + \sin^2 x + \ldots.$ to $\infty = 4 + 2\sqrt{3}, 0 < x < \pi$, then
(A) $x = \frac{\pi}{6}$ (B) $x = \frac{\pi}{3}$
(C) $x = \frac{\pi}{3}$ or $\frac{\pi}{6}$ (D) $x = \frac{\pi}{3}$ or $\frac{2\pi}{3}$

Space for rough work

20. If $a\cos^3\alpha + 3a\cos\alpha\sin^2\alpha = m$ and $a\sin^3\alpha + 3a\cos^2\alpha\sin\alpha = n$, then $(m+n)^{2/3} + (m-n)^{2/3}$ is equal to

(A) $2a^2$ (B) $2a^{1/3}$

(C) $2a^{2/3}$ (D) $2a^3$

21 If $\sin A = n\sin B$, then $\frac{n-1}{n+1}\tan\frac{A+B}{2} =$

(A) $\sin\frac{A-B}{2}$ (B) $\tan\frac{A-B}{2}$

(C) $\cot\frac{A-B}{2}$ (D) None of these

22. If the equation $x^2 + \lambda x + \mu = 0$ has equal roots and one root of the equation $x^2 + \lambda x - 12 = 0$ is 2, then (λ,μ) =

(A) (4, 4) (B) (–4,4)

(C) (4,–4) (D) (–4,–4)

23. If one root of the equation $ax^2 + bx + c = 0$ be n times the other root, then

(A) $na^2 = bc(n+1)^2$ (B) $nb^2 = ac(n+1)^2$

(C) $nc^2 = ab(n+1)^2$ (D) None of these

24. If $\sqrt{3} + i = (a+ib)(c+id)$, then $\tan^{-1}\left(\frac{b}{a}\right) + \tan^{-1}\left(\frac{d}{c}\right)$ has the value

(A) $\frac{\pi}{3} + 2n\pi, n \in I$ (B) $n\pi + \frac{\pi}{6}, n \in I$

(C) $n\pi - \frac{\pi}{3}, n \in I$ (D) $2n\pi - \frac{\pi}{3}, n \in I$

25. The locus of z satisfying the inequality $\log_{1/3}|z+1| > \log_{1/3}|z-1|$ is

(A) $R(z) < 0$ (B) $R(z) > 0$

(C) $I(z) < 0$ (D) None of these

Space for rough work

26. If $n \in N$, then $x^{2n-1} + y^{2n-1}$ is divisible by

(A) $x + y$ (B) $x - y$

(C) $x^2 + y^2$ (D) $x^2 + xy$

27. If n is a natural number then $\left(\frac{n+1}{2}\right)^n \geq n!$ is true when

(A) $n > 1$ (B) $n \geq 1$

(C) $n > 2$ (D) $n \geq 2$

28. From a 60 meter high tower angles of depression of the top and bottom of a house are α and β respectively. If the height of the house is $\frac{60\sin(\beta-\alpha)}{x}$, then x =

(A) $\sin\alpha \sin\beta$ (B) $\cos\alpha \cos\beta$

(C) $\sin\alpha \cos\beta$ (D) $\cos\alpha \sin\beta$

29. Two vertical poles of equal heights are 120 m apart. On the line joining their bottoms, A and B are two points. Angle of elevation of the top of one pole from A is 45° and that of the other pole from B is also 45°. If AB = 30 m, then the height of each pole is

(A) 40 m (B) 45 m

(C) 50 m (D) 42 m

30. If $x = \sqrt{1+\sqrt{1+\sqrt{1+.....\text{to infinity}}}}$, then x =

(A) $\frac{1+\sqrt{5}}{2}$ (B) $\frac{1-\sqrt{5}}{2}$

(C) $\frac{1\pm\sqrt{5}}{2}$ (D) None of these

Space for rough work